Systematic Interrogation of Relationships between the Mitochondrial Genome and Therapeutic Response in Cancer

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Abstract

Tandem advances in high-throughput DNA/RNA sequencing and high-throughput drug screening have facilitated the identification of genome characteristics that render tumors sensitive to specific classes of anti-cancer drugs. These "precision medicine" efforts have almost exclusively focused on the nuclear compartment of the human genome, ignoring the ~16.5 kb mitochondrial chromosome and the 13 mRNAs, 22 tRNAs, and 2 rRNAs encoded by the mitochondrial genome (mtDNA) (**Figure 1**). This is despite the fact that mitochondrial metabolism is important for tumor growth.

Here we propose a strategy to test the hypothesis that certain mitochondrial genome characteristics confer sensitivity to some classes of cancer therapeutics. We present preliminary *in silico* work analyzing drug sensitivity data from >5,000 compounds tested in some 2,400 cancer cell lines, and assessing relationships between these sensitivity measures and mtDNA/mtRNA characteristics.

Introduction

Proteins encoded in mtDNA form key components of four of the five energy-generating complexes collectively known as the electron transport chain (ETC). Mutations in mtDNA have been shown to impair energy production, generate elevated levels of damaging by-products known as reactive oxygen species, and cause severe metabolic disorders. Multiple mtDNA haplotypes may be present in the same cell, tissue, or organism — a condition known as heteroplasmy. MtDNA becomes damaged at a rate an order of magnitude higher than the nuclear genome, which can contribute additional heteroplasmy that may shift over time, sometimes in response to selective pressures (**Figure 2**) and changes in per-cell copy number, which generally ranges from hundreds to thousands.

Multiple studies have demonstrated associations between mtDNA content and response to therapeutics[1-3]. Owing to a number of large public efforts to genomically and pharmacologically characterize hundreds of cancer cell lines, it is now possible to systematically test the relationships between drug response and mtDNA characteristics for thousands of compounds *in silico*.

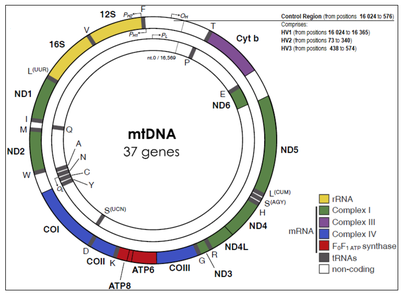

Figure 1. The human mitochondrial genome. From Amorim et al (*PeerJ*, 2019).

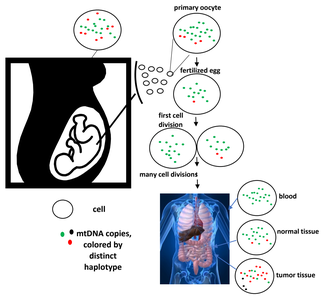

Figure 2. Heteroplasmy shifts can occur temporally and across different tissues within the same individual.

Methods and Materials

We acquired raw DNA/RNA sequence and drug response data (13 sensitivity measures) from the Cancer Cell Line Encyclopedia (CCLE)[4], the Cancer Therapeutic Response Portal (CTRP)[5], Genomics of Drug Sensitivity in Cancer (GDSC)[6], and the PRISM drug repurposing resource[7]. The total data set comprises 2,342 cell lines and 5,322 compounds (**Table 1**).

MtRNA expression levels and mtDNA mutation/heteroplasmy information was extracted from the raw data using custom pipelines established in our laboratory[8,9]. Relationships between drug sensitivity measures and expression levels were assessed using Spearman correlation and those between drug sensitivity measures and mutation status were assessed using Wilcoxon rank-sum test.

Table 1. Data sources and types.

Resource	Sensitivity Measure/Genomic Characteristic	Number of Cell Lines	Number of Compounds
CCLE	IC_{50}	504	24
CCLE	EC_{50}	504	24
CCLE	A_{max}	504	24
CCLE	ActArea	504	24
CTRP	AUC	887	545
GDSC1	AUC	987	367
GDSC1	IC_{50}	987	367
GDSC2	AUC	809	198
GDSC2	IC_{50}	809	198
PRISM	AUC	481	1449
PRISM	IC_{50}	481	1416
PRISM	EC_{50}	481	1449
PRISM	log FC	578	4532
various	mtRNA expression	1368	NA
various	mtDNA mutations/heteroplasmy	1036	NA
Total:		**2342**	**5322**

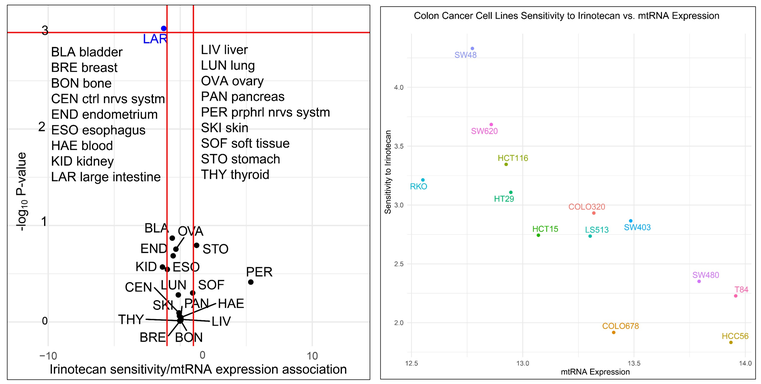

Figure 3. Relationship between Irinotecan sensitivity and mtRNA expression in colon cancer. Within-tissue correlation was strongest for colon (LAR) (left panel). The right panel shows strong negative correlation between mtRNA expression and sensitivity (expression and sensitivity is plotted for all colon cancer cell lines).

Results I: MtRNA expression correlated with drug response in a tissue-dependent manner

As an initial investigation, for each drug we modeled sensitivity as a function of tissue type. Among the strongest associations was an anticorrelation between overall mtRNA expression and sensitivity to Irinotecan (a topoisomerase inhibitor) in colon cancer cell lines (**Figure 3**).

Results II: Truncating mtDNA mutations confer sensitivity

To investigate the impact of mtDNA mutations on drug sensitivity/resistance, we first considered the extreme case of truncating mutations (nonsense and frameshift) in mitochondrial-encoded mRNA genes. Comparing drug responses between: i) cell lines with loss-of-function mutations in these crucial components of the ETC with ii) cell lines wild-type in this regard, we obtained a strong signal from a JNK inhibitor AS601245 (**Figure 4**), which showed significantly stronger efficacy in the mutant cell lines than in the wild-type.

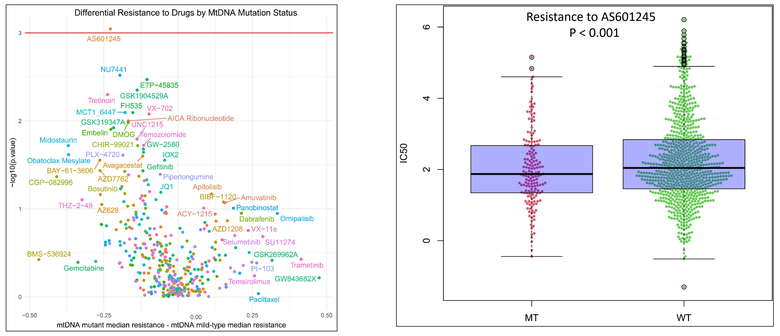

Figure 4. Relationship between drug resistance and loss-of-function mtDNA mutations. Volcano plot (left panel) showing differential resistance of 367 compounds. Right panel shows resistance (as measured by IC_{50}) of each cell line to AS601245.

Experimental Validation (underway)

Given a candidate mtDNA/mtRNA characteristic that is hypothesized to confer sensitivity to specific drug, the hypothesis will be tested using cytoplasmic hybrids ("cybrids"). Briefly, the mitochondria from a sensitive cell line harboring the mtDNA/mtRNA characteristic will be exchanged with mitochondria from a resistant cell line. Both modified cell lines will be treated with the candidate drug to determine whether sensitivity tracks with the mtDNA/mtRNA characteristic as hypothesized **(Figure 5).**

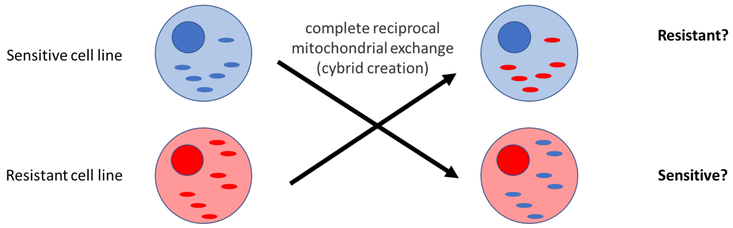

Figure 5. Experimental strategy to validate mitochondrial role. The mitochondria from a sensitive line is swapped with that from a resistant line, and each resulting line is tested for sensitivity.

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